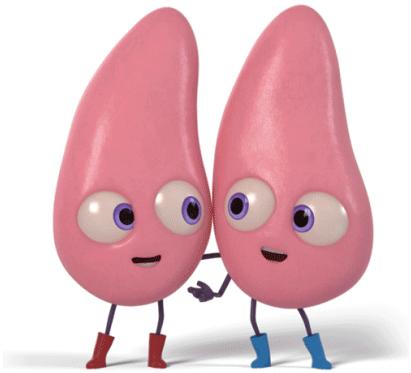

بسم الله الرحمن الرحيم

Pneumonia

By Prof.

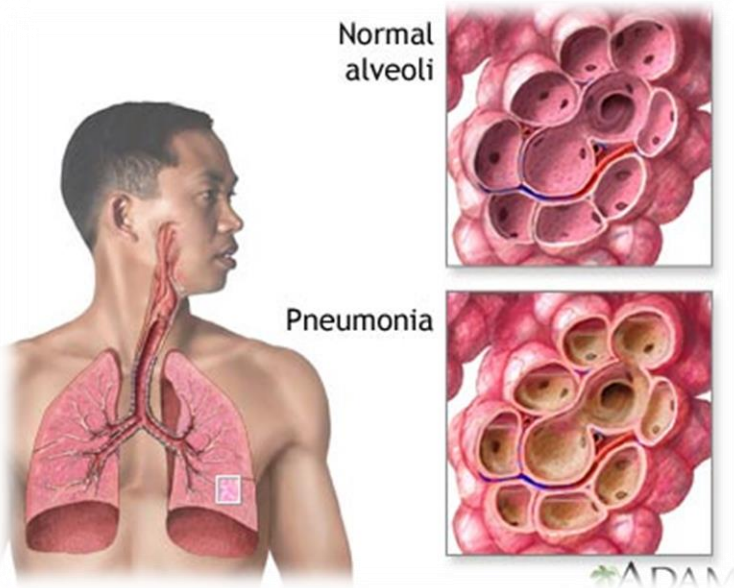
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Pneumonia

Definition:

An inflammation, in general acute, of the lung substance, leading to consolidation of a part or the whole lung due to intra-alveolar exudation.





Pneumonia, a leading cause of death all over the world, it is an infection of the lung parenchyma, most common types and Causes include::

- Community acquired pneumonia(CAP)
  - S. pneumoniae (most common cause)
  - Subsequent causes may vary, but S. pneumoniae is always number 1
- Hospital acquired pneumonia(HAP)
  - Gram-negative bacilli such as E. coli, the other Enterobacteriaceae, or Pseudomonas
  - MRSA

Community Acquired Pneumonia (CAP)

- Community- acquired pneumonia (CAP) refers to pneumonia acquired outside of hospitals or extended- care facilities.
- Internationally CAP is common problem both for community & hospital physicians. It occurs with an annual incidence of about 5-11 per 1000 adult population & rises with age to about 34 per 1000 population for patients aged over 75 years

- The annual incidence of patients that require hospital admission varies from approximately 1-4 per 100000 population
- The morality rate are low (<1%) for patients managed in the community, higher in patients admitted to hospital around 5-12% and highest for patients requiring mechanical ventilation with rates from around 35% to 50%
- It is thus a common disease & can have a considerable impact on health care resources

Community acquired pneumonia(CAP) Causes include:

"Typical"	
Strep. pneumoniae	15-35%
Haemophilus	2-10%
Moraxella	5%
"Atypical"	
Legionella	0-15%
Mycoplasma	10%
Chlamydia	5-10%
Viral	2-20%
Unknown	30-60%

- Specific predispositions are as follows:
  - Cigarette smoking,
  - Diabetes,
  - Alcoholism,
  - Malnutrition,
  - Obstruction of the bronchi from tumors,
  - Immunosuppression in general

Clinical Presentation



- Patients with pneumonia present with cough, fever, and often sputum production. Severe pneumonia of any cause may present with dyspnea.
- The quality and degree of sputum produced might provide useful clues to the microbiologic etiology of pneumonia at the initial presentation.
- Bacterial infections such as S. pneumoniae, Haemophilus, and Klebsiella have significant purulent sputum production because they are infections of the alveolar air space.

- The sputum with S. pneumoniae is described as **rusty**. The “rust” is simply hemoptysis. As the blood oxidizes, it becomes brownish-red color
- Interstitial infections such as those caused by viruses, Mycoplasma, and sometimes Legionella often give a **non-productive or “dry” cough**.

- Any cause of pneumonia may be associated with pleuritic chest pain. This is pain worsened by inspiration.
- Commonly, **pleuritic pain** is associated with lobar pneumonia, such as that caused by Pneumococcus. This is because of localized inflammation of the pleura by the infection.
- Lobar pneumonia is the type most commonly associated with signs of **consolidation** on examination.

On physical examination pneumonia presents with :

1. **Inspection: ↓ movement sudden breath catch**
2. **Palpation:**
   - **↑ T.V.F.**
   - **Central mediastinum**
   - **Palpable friction rub**
3. **Percussion:**
   - **Impaired note → dull note**
4. **Auscultation:**
   - **Bronchial breathing "high pitch"**
   - **↑ Whispering & ↑ vocal resonance**
   - **Crepitations (medium sized, late inspiratory).**
   - **Pleural rub**

Diagnosis:

I-Chest x-ray is The most important initial test for pneumonia, which both shows the presence of disease and gives the initial clues to determining the diagnosis.

- Infiltrates are either
  - localized to a single lobe **or** the whole lung
  - bilateral and interstitial
- It is the most important clue.

- S. pneumoniae (and other causes of “typical” pneumonia) usually appear as a **lobar pneumonia**, sometimes with parapneumonic pleural effusion.

- **Interstitial infiltrates** are associated with viral, Mycoplasma, Chlamydia, Coxiella, and sometimes Legionella pneumoniae.

II. Hematology:

- Neutrophil leucocytosis is usual in pneumococcal pneumonia & other bacterial pneumonias. The total white blood cell count doesn't often exceed **30,000** & is more frequently **12,000 – 15,000/mm^3**
- Lymphopenia in legionella & viral pneumonia
- Sometimes there is eosinphilia in convalescence
- Normocytic anemia commonly seen in severe pneumonia.
- The erythrocyte sedimentation rate is usually raised & may be very high, a rate of **100mm** or more isn't uncommon. It decreases in convalescence.
- Increased Inflammatory markers as c-reactive protein
- Raised procalcitonin level in bacterial infection

III-Sputum sample should be obtained for both Gram stain as well as culture. Sputum culture is the most specific diagnostic test for lobar pneumonia, such as with S. pneumoniae, Staphylococcus, Klebsiella, and Haemophilus.

- The other “atypical” organisms (viral, Mycoplasma, Chlamydia, Coxiella, etc.) will not show up on Gram stain or regular bacterial culture for various reasons, so serology must be done to detect them.

The diagnosis of CAP require :

(1)Basic finding → a new pulmonary infiltrate seen on a chest radiograph that is obtained within 24 h of presentation;

(2) at least one of the major criteria →

cough, sputum production, or temperature≥37.8°C.

or at least two of the minor criteria →

- **pleuritic chest pain, dyspnea,**
- **altered mental status,**
- **pulmonary consolidation, and WBC ≥ 12,000 cells/L.**

Treatment of CAP



- General lines
- Antibiotic therapy

***General* lines:**

- Treatment of predisposing factors
- Bed rest & proper nursing; easily digested food & excess Fluid
- Cough sedatives early; then expectorants
- Supportive measures : vitamin A, B & C
- Analgesics to control pleurtic pain.
- Oxygen therapy.

Antibiotic therapy for CAP should:

- Started empirically
- Selected with patients characteristics, place of acquisition and severity of the disease
- When specific pathogen is identified ; specific antibiotic could be used

Empiric Therapy of Community-Acquired Pneumonia

Outpatient treatment

- Young, healthy patients with no comorbidities or antibiotic use in preceding 3 months:
  - Azithromycin, clarithromycin, or doxycycline
- With comorbidities, or antibiotic use in preceding 3 months use either
  - Respiratory fluoroquinolones (Levofloxacin, moxifloxacin)
  - **or** oral beta-lactam (Oral second- and third-generation cephalosporins or amoxicillin/clavulanate) + a macrolide or doxycycline

Hospitalized patients

- Respiratory fluoroquinolones (levofloxacin, or moxifloxacin,)

Or

- Second- or third-generation cephalosporins (cefuroxime or ceftriaxone) combined with a macrolide or doxycycline

Or

- Beta-lactam/beta-lactamase combination drug (ampicillin/sulbactam; ticarcillin/ clavulanate; piperacillin/tazobactam) combined with doxycycline or a macrolide

- If risk factors for Community acquired -MRSA, use above therapy + vancomycin or linezolid or ceftaroline + a macrolide or doxycycline

- If risk factors for Pseudomonas, use antipseudomonal beta-lactam (like cefepime or meropenem) + ciprofloxacin; for penicillin-allergy, use a respiratory quinolone + aztreonam

Duration of therapy

- Patients with CAP should be treated with antibiotics for a minimum of 5 days ,however 7-10 days is usually adequate

- Should be afebrile for 3-4 days and have no more than one CAP-associated sign of clinical instability prior to discontinuation.

- Patients who have been hospitalized and are receiving intravenous antibiotics should be switched to oral therapy as soon as they are
  - hemodynamically stable,
  - improving clinically,
  - able to ingest medications, and
  - have normal digestion

- Chest x-ray should be done every week
- Usually the x-ray film return to normal within 2-6 weeks however in old age patients ;it may take longer duration
- If resolution delayed we must search for underlying cause

Causes of delayed resolving pneumonia:

- 1- inadequate ttt (type, dose, or time)
- 2-Specific pathogen,eg;TB
- 3-Underlying pathology(neoplasia, foreign body, bronchiectesis)
- 4-Empyema
- 5-Immunocompromised: DM, CHF, alcoholism,etc.

Hospital acquired pneumonia

The 2016 Infectious Diseases Society of America (IDSA)/American Thoracic Society (ATS) guidelines distinguish the following types of pneumonia:

- ●**Hospital-acquired (or nosocomial) pneumonia (HAP)** is pneumonia that occurs 48 hours or more after admission and did not appear to be incubating at the time of admission.
- ●**Ventilator-associated pneumonia (VAP)** is a type of HAP that develops more than 48 to 72 hours after endotracheal intubation.

Treatment of HAP

- Regardless of severity, the initial empirical treatment regimens for HAP who have early-onset HAP (within 4 to 5 days of hospitalization) should include coverage for a group of core organisms that includes
  - Antibiotic-sensitive, aerobic, enteric gram-negative bacilli **(*e.g.*, *E. coli*, *Klebsiella* spp., *Proteus* spp., and *Serratia marcescens*),**
  - community pathogens such as ***Haemophilus influenzae***
  - ***Streptococcus pneumoniae***, as well as methicillin-sensitive ***S. aureus***

Early onset HAP Treatment

Extended spectrum cephalosporin
Or
B-lactam+b-lactamase inhibitor
Or
fluoroquinolones

- Those patients who develop pneumonia **late-onset** after 5–7 days in the hospital are at increased risk of infection from:
  - drug-resistant, gram-negative bacilli (Pseudomonas, Klebsiella, E. coli, *Acinetobacter* spp..) or gram-positive bacilli such as MRSA.

- Third-generation cephalosporin with antipseudomonal activity (e.g., ceftazidime) or carbapenem (e.g., imipenem)
- Beta-lactam/beta-lactamase inhibitor combinations (such as piperacillin/tazobactam) plus an ***antipseudomonal fluoroquinolone*** (e.g., ciprofloxacin) and vancomycin or linezolid to cover MRSA.
- Possible addition of polymyxin(colistin) or an aminoglycoside (gentamicin, tobramycin, amikacin) to empiric gram-negative coverage



- Pneumonia is a wild monster as even with modern medical care the case mortality is about 12%.
- Prevention through Smoking cessation, Pneumococcal & Influenza vaccination must be the main goal in pneumonia management
- Early antibiotic administration within 4-6 hours
- Empiric antibiotic treatment as per guidelines (IDSA / ATS) must be given
- Typical and Atypical pathogens must be considered in the choice of antibiotic therapy.





Thank You

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